GlobalMart End-to-End Lakehouse & Generative AI Architecture

1. Architecture Overview (Purpose & Philosophy)

GlobalMart's architecture is designed to move the organization from **raw data ingestion** to **actionable intelligence**, using a governed Lakehouse foundation enhanced with Generative AI.

The solution follows the **Databricks Medallion Architecture** (Bronze → Silver → Gold) and extends it with a **GenAI Intelligence Layer**. Each layer has a clear responsibility, and data only moves forward once it meets the quality and semantic requirements of the next layer.

The architecture directly addresses three critical business failures:

- Revenue audit inconsistencies caused by duplicate and invalid records
- Return fraud losses caused by fragmented regional visibility
- Inventory blindspots caused by delayed, analyst-dependent reporting

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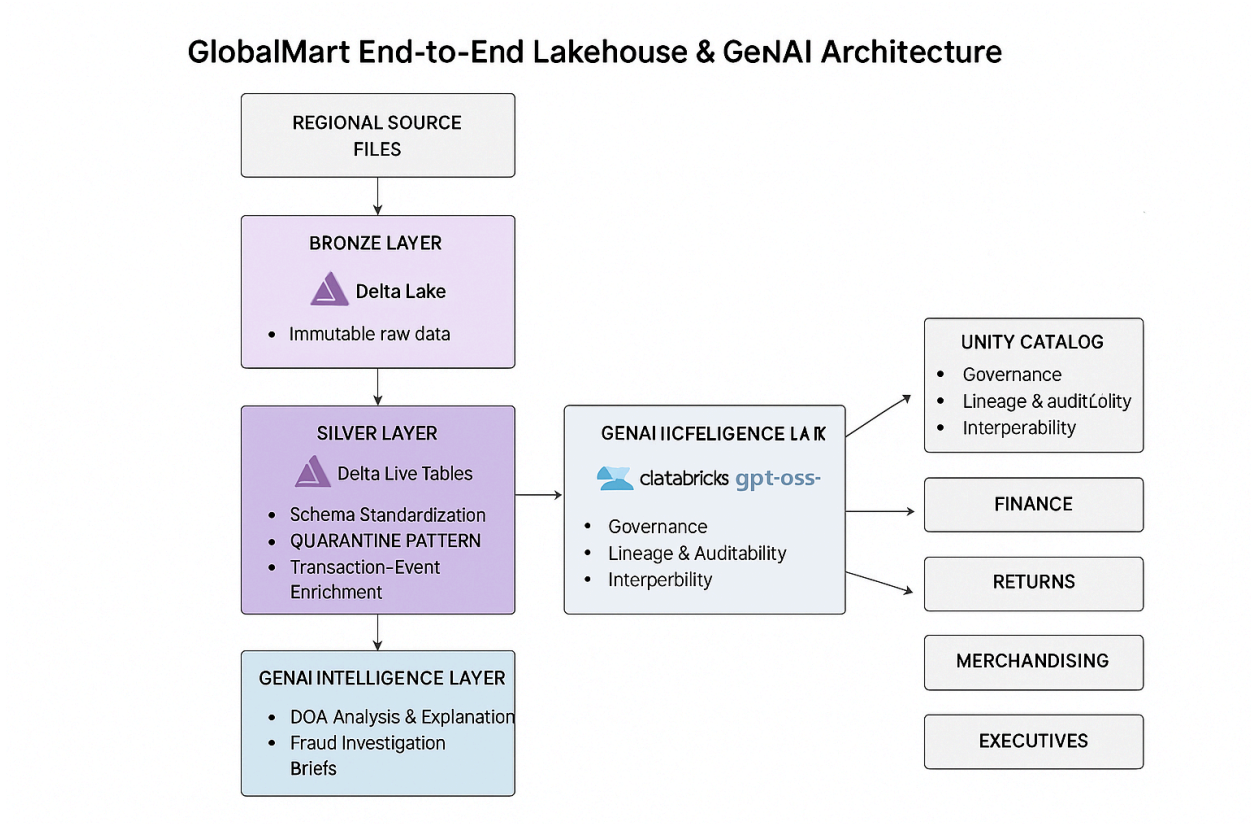
2. Data Flow Diagram (End-to-End)

End-to-End Flow Description

```
None
Regional Source Files (6 regions)
        ↓
Auto Loader (Bronze Layer)
        ↓
Delta Live Tables — Silver Layer
   - Standardization
   - Data Quality Rules
   - Quarantine Handling
        ↓
Delta Live Tables — Gold Layer
```

```
   - Fact Tables
   - Dimension Tables
   - Pre-Aggregated Metrics
   - Fraud Investigation Outputs
        ↓
Generative AI Intelligence Layer
        ↓
Business Users
(Finance, Returns, Merchandising, Executives)
```



Description

- Raw operational files are ingested incrementally
- Data is cleaned, validated, and enriched
- Business-ready analytics are produced

- GenAI converts metrics into plain-English explanations
- Outputs are consumed via tables, dashboards, and narratives

3. Component Map (Tools, Layer, Role)

Ingestion & Storage

- **Auto Loader**
  Layer: Bronze
  Role: Incremental ingestion of raw files, schema drift handling
- **Delta Lake**
  Layer: All
  Role: Reliable, ACID-compliant storage

Processing & Transformation

- **Delta Live Tables (DLT)**
  Layer: Silver & Gold
  Role: Streaming transformations, data quality enforcement, incremental processing

Governance

- **Unity Catalog**
  Layer: Cross-cutting
  Role: Centralized metadata, schema governance, access control, lineage

Analytics & Intelligence

- **SQL Warehouses / Notebooks**
  Layer: Consumption
  Role: Querying, dashboards, validation
- **databricks-gpt-oss-20b**
  Layer: GenAI
  Role: Natural-language explanations and investigation briefs

4. Data Quality Strategy (What, Where, Why)

Bronze Layer

- No validation or transformation
- Raw data preserved exactly as received
- Enables replay and audit traceability

Silver Layer (Primary Trust Boundary)

Data quality rules enforced include:

- Required identifiers (customer_id, order_id)
- Valid categorical values (segments, ship modes)
- Non-negative financial values
- Referential integrity checks

Quarantine Pattern

- Invalid records are **never dropped**
- Rejected rows are written to dedicated quarantine tables
- Each rejected record includes a clear rejection reason
- Enables transparency for finance and audit teams

Gold Layer

- Consumes **only validated Silver data**
- Guarantees analytical correctness
- No business logic operates on rejected records

5. Silver Layer Key Design Decisions

Unified Customer Identity

- Regional customer variants are normalized into a single customer representation
- Eliminates duplicate attribution in revenue and returns analysis

Event-Time Region Propagation

- Region is copied into orders during Silver processing
- Region becomes a property of the **transaction event**, not the customer
- Prevents historical distortion if customer attributes change later
- Crucial for revenue audit accuracy and fraud analytics

6. Analytics Layer Structure (Gold Layer)

Fact Tables

- **fact_sales**
  - One row per completed sales event
  - Measures revenue, quantities, monetary values
- **fact_returns**
  - One row per return event
  - Tracks refunds and return volume

Dimension Tables

- Customers
- Products
- Vendors

Dimensions provide descriptive context only and do not contain metrics.

Pre-Aggregated Gold Tables

Created to support frequent business queries without scanning raw facts:

- Monthly revenue by region
- Vendor return rates
- Customer return profiles
- Slow-moving products

These tables stabilize metrics and eliminate repeated analyst intervention.

7. Fraud Detection Architecture (Deterministic Analytics)

Why Rule-Based (Not ML)

- Fraud detection must be explainable
- Thresholds must be auditable
- Outputs must be defensible to finance and compliance teams

Anomaly Scoring System

Each customer receives a **0–100 anomaly score** based on five independent rules:

- High return volume
- High refund value
- High return-to-order ratio
- Recent spikes in return activity
- Refund value exceeding purchase value

Rules are weighted and combined deterministically.

Output

- A Gold table (`flagged_return_customers`)
- One row per flagged customer
- Explicit violation list
- Stable investigation queue for operations

An empty table is a valid outcome if no customer crosses the confidence threshold.

8. Generative AI Integration

Role of GenAI

GenAI is used **only as an explainer**, never as a classifier or decision engine.

Inputs

- Curated Gold tables
- Fraud scores and rule violations
- Quarantine summaries

Outputs

- Audit-ready data quality narratives
- Fraud investigation briefs
- Business-friendly explanations

All GenAI outputs are **persisted back to the Gold layer** for governance and auditability.

Implemented Use Cases

- AI Data Quality Reporter
- Returns Fraud Investigator

9. Governance, Auditability & Trust

Governance

- Unity Catalog tracks lineage end-to-end
- Fine-grained access control across teams
- All AI outputs are discoverable and reviewable

Audit Readiness

- Every exclusion is explainable
- Every fraud flag is justified with data
- No black-box decisions

10. Business Impact Summary

- Finance teams gain audit-ready explanations
- Returns teams receive a prioritized investigation queue
- Merchandising teams act proactively on trends
- Executives receive synthesized intelligence, not dashboards

Final Closing Statement

This architecture demonstrates how GlobalMart moves from **data pipelines** to a **decision platform**, where governed analytics and Generative AI work together to produce trusted, explainable, and actionable intelligence.